SeungJu Cha

Ph.D. Student in Artificial Intelligence
Hanyang University, Republic of Korea

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Research Interests

- **Diffusion / Flow Models:** Prompt Alignment, Efficient Generation, Physics-Aware Video Generation.
- **Autoregressive Video Diffusion:** Long-Horizon Video Generation, Temporal Consistency in World Models.

Education

Hanyang University, Republic of Korea 2026 – Present
Ph.D. Student, Department of Artificial Intelligence
Advisor: Prof. Dong-Jin Kim

Hanyang University, Republic of Korea 2024 – 2026
M.S., Department of Artificial Intelligence
Advisor: Prof. Dong-Jin Kim

Hanyang University, Republic of Korea 2018 – 2024
B.S., Mechanical Engineering

Publications

Generative Modeling

- **Which Tokens to Merge? Diffusion Dynamics for Efficient Image Generation**
  SeungJu Cha, Ye-Chan Kim, HyunGee Kim, Sungho Koh, Dong-Jin Kim
  NeurIPS 2026 [Link]
- **ScaleDiff: Higher-Resolution Image Synthesis via Efficient and Model-Agnostic Diffusion**
  SungHo Koh, **SeungJu Cha**, Hyunwoo Oh, Kwanyoung Lee, Dong-Jin Kim
  NeurIPS 2025 [Link]
- **Beyond Pairwise Repulsion: Collective Diversity via Determinant Maximization in Diffusion Sampling**
  Kwanyoung Lee, **SeungJu Cha**, Yebin Ahn, Sungho Koh, Dong-Jin Kim
  ACCV 2026 (Oral)
- **VerbDiff: Text-Only Diffusion Models with Enhanced Interaction Awareness**
  SeungJu Cha, Kwanyoung Lee, Youngchan Kim, Hyunwoo Oh, Dong-Jin Kim
  CVPR 2025 [Link]
- **Not Just a Subject: Capturing Adaptive Scene-Specific Variance for Visual Storytelling**
  SeungJu Cha, Ye-Chan Kim, Kwanyoung Lee, Dong-Jin Kim
  ACCV 2026
- **Adaptive Auxiliary Prompt Blending for Target-Faithful Diffusion Generation**
  Kwanyoung Lee, **SeungJu Cha**, Yebin Ahn, Hyunwoo Oh, Sungho Koh, Dong-Jin Kim
  CVPR 2026 [Link]
- **ADAPT: Attention Driven Adaptive Prompt Scheduling and InTerpolating Orthogonal Complements for Rare Concepts Generation**
  Kwanyoung Lee, Hyunwoo Oh, **SeungJu Cha**, Sungho Koh, Dong-Jin Kim
  CVPR 2026 Findings [Link]
- **CatchPhrase: EXPrompt-Guided Encoder Adaptation for Audio-to-Image Generation**
  Hyunwoo Oh, **SeungJu Cha**, Kwanyoung Lee, Si-Woo Kim, Dong-Jin Kim
  ACM Multimedia (ACM MM) 2025 [Link]

Other Publications

- **SAIL: Similarity-Aware Guidance and Inter-Caption Augmentation-based Learning for Weakly-Supervised Dense Video Captioning**
  Ye-Chan Kim, **SeungJu Cha**, Si-Woo Kim, MinJu Jeon, HyunGee Kim, Dong-Jin Kim

CVPR 2026 [Link]

- **Seeing Before Synthesizing: VLM-Guided Transition Event Discovery for Weakly-Supervised Dense Video Captioning**
  Ye-Chan Kim, Seunghee Choi, **SeungJu Cha**, Si-Woo Kim, Hwiseon Kim, Hyungee Kim, Dong-Jin Kim
  EMNLP 2026 [Link]
- **ADuLTS: Appearance Descriptions under Long-Tailed Scenarios with Diverse Synthesized Images**
  SeungJu Cha, Seunghee Choi, Kwanyoung Lee, Dong-Jin Kim
  Computer Vision and Image Understanding (CVIU), 2026 [Link]
- **Sida: Synthetic Image Driven Zero-Shot Domain Adaptation**
  Youngchan Kim, **SeungJu Cha**, Si-Woo Kim, Taehwan Kim, Dong-Jin Kim
  ACM Multimedia (ACM MM) 2025 [Link]

Technical Skills

Programming & Frameworks: Python, PyTorch
Development Tools: Linux, Git

Academic Service

Reviewer: AAAI 2026, NeurIPS 2026 Top Reviewer, ACCV 2026, AAAI 2027